

- **strictly dom:** $u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i}) \forall s_{-i} \in S_{-i}$.
- **best-response:** $R_i(s_{-i}) := \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})$.
- **Nash:** $s_i^* \in R_i(s_{-i}^*)$ iff $u_i(s_i^*, s_{-i}^*) = \max u_i(s_i, s_{-i}^*)$.
- $G(R_i) := \{(s_i, s_{-i}) : s_i \in R_i(s_{-i}), s_{-i} \in S_{-i}\}$: **graph** of R_i , $\bigcap_{i=1}^n G(R_i)$ are exactly all the Nash equilibria.
- If $(s_1^*, s_2^*, \dots, s_n^*)$ are a Nash equilibrium, then no s_i can be eliminated by IESDS. **Proof: suppose s_j is the first to be eliminated.**
- If IESDS results in a single set of strategies, they are the unique Nash equilibrium of the game. **Proof: uniqueness is trivial. For existence, suppose $s_i^* \notin R_i(s_{-i}^*) \neq \emptyset$ for some i .**
- **Mixed:** $p_i := (p_{ij})_{j=1}^k$ s.t. $P(i \text{ plays } s_{ij}) = p_{ij}$.
- **$\mathbb{E}$ payoff (pure):** $v_i(x, p_j) := \sum_{n=1}^N p_{jn} u_i(x, s_{jn})$
- **Expected payoff for mixed strategy:** $v_i(p_i, p_j) := \sum_{m=1}^M p_{im} \sum_{n=1}^N p_{jn} u_i(s_{im}, s_{jn})$
- If s_{ik} is eliminated by IESDS, then $p_{ik}^* = 0$.
- **Nash's Theorem:** at least one Nash equilibrium exists for any finite game if mixed strategies are allowed.
- **Market clearing price:** $D(P) = Q(P)$.
- **Reservation price:** $\min\{a : D(a) = 0\}$
- **Cournot Model of Duopoly:**
 - Demand: $Q(P) = ka - kP$
 - $P(Q) = \{a - \frac{1}{k}Q\} \mathbf{1}\{Q < ka\}$.
 - $\pi_i(q_i, q_j) = \begin{cases} (a - \frac{1}{k}q_i - \frac{1}{k}q_j - c)q_i & \text{if } Q < ka \\ -cq_i & \text{if } Q \geq ka \end{cases}$
 - $R_i(q_j) = \frac{1}{2}(ka - q_j - kc) \mathbf{1}\{q_j < k(a - c)\}$.
- **Bertrand Model of Duopoly:**
 - Demand: $q_i(p_i, p_j) = ka - kp_i + bp_j$
 - Profit: $\pi_i(p_i, p_j) = (ka - kp_i + bp_j)(p_i - c)$.
 - Best response: $R_i(p_j) = \frac{1}{2k}(ka + bp_j + kc)$.
 - Nash eq: $p_1^* = p_2^* = \frac{ka+kc}{2k-b}$, i.e., $b > 2k$.
 - Homogeneous prod Nash eq: $p_1^* = p_2^* = c$ ($b = 0$)
- **Problem of the commons:**
 - Unit util | total consumption: $v(x)$ **strictly concave** for $x < X_{\max}$ and $v(x) = 0$ for $x \geq X_{\max}$.

- Payoff: $u_i(x_i, x_{-i}) := x_i v(X) - cx_i$.
- Nash eq solves $v(X^*) + x_i^* \frac{dv}{dx_i} \Big|_{x_i=x_i^*} - c = 0$.
- If $v(x) = a - x^2$ for $x < X_{\max}$, then $x_i^* = \sqrt{\frac{a-c}{n^2+2n}}$.
- Social welfare: $w(X) := Xv(X) - Xc$.
- If x^* optimises w , then $x^* < X^*$. **Proof: if $x^* \geq X^*$, then $v(X^*) \geq v(x^*)$ and $0 > v'(X^*) \geq v'(x^*)$. Use this to show $v(X^*) + \frac{X^*}{n}v'(X^*) > v(x^*) + x^*v'(x^*)$. Use stationary points to show $w'(x^*) = 0$ and $u'_i(x_i^*) = 0$.**

• **Final-offer arbitration:**

- Union offers w_u and firm offers w_f .
- Arbitrator expectation $x \sim f$ with median x_m .
- Expected wage: $\phi(w_f, w_u) = \mathbb{E}[w] = w_f F\left(\frac{w_f + w_u}{2}\right) + w_u \left(1 - F\left(\frac{w_f + w_u}{2}\right)\right)$.
- $R_f(w_u) = \arg\min_{0 \leq w_f < w_u^*} \phi(w_f, w_u^*)$
- $R_u(w_f) = \arg\max_{w_u > w_f^*} \phi(w_f^*, w_u)$
- Nash eq: $w_f^* = x_m - \frac{1}{2f(x_m)}$, $w_u^* = x_m + \frac{1}{2f(x_m)}$.

• **Backward induction:**

- Define $b_j(b_1) := R_j(b_1, \dots, b_{j-1})$ as player j 's best action given player 1's action.
- $a_i^* := \arg\max_{a_i \in A_i} u_1(a_1, b_2(a_1), \dots, b_n(a_1))$ and $a_i^* := b_i(a_1^*)$.

• **Extensive-form game:** players + when a move is taken + strategy spaces + knowledges + payoffs of all action sequences.

• **Info set:** a set of **decision nodes** joined with **dotted line in a game tree** for the same player at a turn s.t. the player cannot deduce the predecessor. **Perfect information** $\iff$ all info sets are singletons.

• **Subgame:** a game based on a subtree of a game tree s.t. its decision nodes can be partitioned by info sets.

• **Strategy in dynamic games:** $s_i: \prod_{j=1}^{i-1} A_j \rightarrow A_i$.

• Action of i given strategies $s := (s_j)_{j=1}^n$ of all players is $a_i(s) = s_i(a_1(s), \dots, a_{i-1}(s))$.

• **Dyn game payoff:** $\tilde{u}(s) := u(a_1(s), \dots, a_n(s))$

• $(a_1^*, R_2(a_1^*))$ is backward induction outcome $\implies (a_1^*, R_2)$ is Nash eq. **Converse false in general.**

• **Subgame perfect Nash equilibrium** is a Nash equilibrium in every subgame.

• **Stackelberg Model:** Dominant 1 and follower 2.

- Price: $P(Q) = (a - \frac{1}{k}Q) \mathbf{1}\{Q < ka\}$.
- $R_2(q_1) = \frac{1}{2}(ka - q_1 - kc) \mathbf{1}\{q_1 < k(a - c)\}$.

	Cournot	Stackelberg
(q_1^*, q_2^*)	$(\frac{ka-kc}{3}, \frac{ka-kc}{3})$	$(\frac{ka-kc}{2}, \frac{ka-kc}{4})$
P	$c + \frac{a-c}{3}$	$c + \frac{a-c}{4}$
(π_1^*, π_2^*)	$(\frac{k^2(a-c)^2}{9}, \frac{k^2(a-c)^2}{9})$	$(\frac{k^2(a-c)^2}{8}, \frac{k^2(a-c)^2}{16})$
$\pi_1^* + \pi_2^*$	$\frac{2k^2(a-c)^2}{9}$	$\frac{3k^2(a-c)^2}{16}$

• **Three-period bargaining:** if receive (u_1, u_2) in period $i + 1$, then player i offer δu_j and player j accept.

• In 3-period bargaining, 1 offer $s_1(1) = 1 - \delta(1 - \delta\bar{s}_1)$, $s_2(1) = \delta(1 - \delta\bar{s}_1)$ and 2 accept.

• **Infinite-horizon bargaining:** if $(\bar{s}_1, \bar{s}_2)$ are final offers received in backward induction, then by collapsing stage 3 onwards, the game is equivalent to a 3-period bargaining with $\begin{cases} \bar{s}_1 = 1 - \delta(1 - \delta\bar{s}_1) \\ \bar{s}_2 = \delta(1 - \delta\bar{s}_1) \end{cases}$ which yields $\bar{s}_1 = \frac{1}{1+\delta}$, $\bar{s}_2 = \frac{\delta}{1+\delta}$.

• **Multi-stage game:** dynamic games which can be partitioned into static stage games. The game is **repeated** if every stage is the same.

• If the stage game of a repeated game has a unique Nash equilibrium, then the repeated game has a unique subgame perfect Nash equilibrium.

• **Present value:** $\sum_{i=1}^{\infty} \delta^{i-1} \pi_i$ where π_i is the stage i payoff and $\delta \in (0, 1)$ is the **discount factor**.

• If the stage game with left (not cooperate) and right (cooperate) strategies is $\begin{bmatrix} L & R \\ L & (1,1) & (5,0) \\ R & (0,5) & (4,4) \end{bmatrix}$, then (NC, NC) is always a Nash equilibrium and (T, T) is a Nash equilibrium iff $\delta \geq \frac{1}{4}$. **Proof: show that T is the best response for T , i.e., if player 1 plays T , then player 2 will end up playing T .**

• **Asymmetric info Cournot:** c_1 is known, $c_2 = c_H$ with prob. θ and c_L with prob. $\rho = 1 - \theta$.

- $q_1^* = \frac{a-2c+\theta c_H+\rho c_L}{3}$, $q_2^* = \frac{a-2c_X+c}{3} + \frac{p_Y}{6} (c_H - c_L)$.
- $q_2^*(c_L) > q_2^*(c_H)$, $\pi_L = q_2(c_L)^2 > q_2(c_H)^2 = \pi_H$

- **Bayesian game:** each player has belief $P_i(t_{-i} | t_i)$ on others' types given self type, $P_i(t_{-i} | t_i) = \frac{P_i(t_{-i}, t_i)}{\sum P_i(t_{-i}, t_i)}$.
- **Bayesian game strategy:** $s_i: T_i \rightarrow A_i$.
- $s^*(t_i) = \operatorname{argmax}_{a_i \in A_i} \mathbb{E}_{t_{-i}} [u_i(a_i, s_{-i}^*(t_{-i}); t_i)]$
- A Bayesian game can be viewed as a collection G of possible games. A type space T_i is a partition of G with $\sigma_i: G \rightarrow T_i$ as the quotient map.
- Given player i 's type, $P(g | t_i) = \frac{P(g)}{P(t_i)} = \frac{P(g)}{\sum_{g_i \in t_i} P(g_i)}$ if $g \in t_i$ and 0 otherwise. Given strategy profile s_{-i} , $u_i(a_i, s_{-i}; t_i) = \frac{1}{P(t_i)} \sum_{g \in t_i} P(g) u_i(a_i, s_{-i}(\sigma_{-i}(g)))$.
- **1st-Price Sealed-Bid:** valuation $v_i \sim U(0, 1)$ as type, $u_i = v_i - b_i$ if $b_i > b_j$, $\frac{v_i - b_i}{2}$ if $b_i = b_j$ and 0 otherwise. Nash eq is linear func $b_i(v_i) = a_i + c_i v_i = \frac{v_i + a_j}{2}$.
- **2nd-Price Sealed-Bid:** bid at valuation is Nash eq as weakly dominate $u_i(v_i, r_i; v_i) = \max\{v_i - r_i, 0\}$.
- **Signal game:**
 1. Nature draws t_i s.t. $P(\tilde{t} = t_i) = P(t_i)$ for Sender.
 2. Sender sends $m_i \in M$
 3. Receiver observes m_i only and chooses $a_i \in A$
 4. $U_S(t_i, m_i, a_i)$ and $U_R(t_i, m_i, a_i)$ are payoffs.
 5. $a^*(m) = \operatorname{argmax}_{a \in A} \sum_{t \in T} \mu(t | m) U_R(t, m, a)$, $m^*(t) = \operatorname{argmax}_{m \in M} U_S(t, m, a^*(m))$
 6. $\mu(t | m) = \frac{P(t)}{\sum_{t' \in T_m} P(t')}$, where T_m is all types s.t. $m^*(t) = m$. $T_m = \emptyset \implies \mu$ anything.
 7. $\mu(t | m) = P(\tilde{t} = t | m^*(\tilde{t}) = m)$.
 8. (m, a, μ) satisfy 5 and 6 is a perfect Bayesian eq
 9. strategies — (x, x) : pooling, (x, y) : separation
- Finding PBE: check only Nash eq
- **Job-Market signal game:**
 - $\eta \in \{\eta_H, \eta_L\}$: **ability** of worker (type)
 - $e \in \{e_c, e_s\}$: **education level** of worker (msg)
 - $w \in [0, \infty)$: **wage** of worker (action)
 - $c(\eta, e) = c_1(\eta) + c_2(e)$: cost to obtain education
 - $U_W(\eta, e, w) = w - c(\eta, e)$: worker payoff
 - $U_F(\eta, e, w) = -(y(\eta, e) - w)^2$: firm payoff (y is worker output)
 - $\Delta(\eta) = U_W(\eta, e_c, w^*(e_c)) - U_W(\eta, e_s, w^*(e_s))$ ind. of η . $\Delta > 0 \implies e_c$ better $\implies$ separation is not optimal

- Pooling: if $P(\eta_H) = r$, $w^*(e_i, r) = ry(\eta_H, e_i) + (1-r)y(\eta_L, e_i) = \mathbb{E}[y]$ is optimal for firm
- **(2-player) Co-operative game:**
 - $H \subseteq \mathbb{R}^2$: **attainable payoffs**
 - $d \in H$: **threat pt** — achieved when fail to trade
 - $u \in H \iff \exists$ action of both players to attain u
- (u, v) **dominates** $(u', v') \iff u \geq u', v \geq v'$. **Pareto optimal** $\iff$ dominated by nothing
- **2-person bargaining** (H, d) :
 - $H \subseteq \mathbb{R}^2$ is compact and convex, $d \in H$
 - H contains some u dominating d
 - W : set of all w-person bargaining games
- **Nash bargaining soln:** $f: W \rightarrow \mathbb{R}^2$ s.t.
 1. $f(H, d) \in H$ (feasible)
 2. $f(H, d) \geq d$ (individual rational)
 3. $f(H, d)$ is Pareto optimal
 4. if $H' = \{(a_x x + b_x, a_y y + b_y) | (x, y) \in H\}$, then $f_i(H', d') = a_i f_i(H, d) + b_i$ (invariant under linear transformation)
 5. if $d_1 = d_2$ and $(x, y) \in H \iff (y, x) \in H$, then $f_1(H, d_1) = f_2(H, d_2)$ (symmetry)
 6. if $d' = d$ and $H \subseteq H'$, then $f(H', d') = f(H, d)$ (independence of irrelevant alternatives)
- Every $(H, d) \in W$ has a unique Nash bargaining soln u^* s.t. $(u_1^* - d_1)(u_2^* - d_2) > (u_1 - d_1)(u_2 - d_2)$ for all $u^* \neq u \geq d$, i.e., u^* is global maximum of the product
- **coalition:** non-empty subset of set of all players
- **char func:** $v(S)$ is the amount surely attainable by S s.t. $v(\emptyset) = 0$ and $v(A \cup B) \geq v(A) + v(B) \forall$ disjoint A, B (super-additivity)
- **imputation:** a payoff vec x s.t. $\sum x_i = v(N)$ (grp rational) and $x_i \geq v(\{i\})$ (individually rational). $I(N, v)$ set of all imputations of game (N, v)
- $x, y \in I(N, v)$, x **dominates** y via S ($x \succ_S y$) if $x_i > y_i \forall i \in S$ and $\sum_{i \in S} x_i \leq v(S)$. x **dominates** y if $\exists S$ s.t. $x \succ_S y$.
- **core** $C(N, v)$: undominated imputations. $C(N, v)$ is all payoff vec s.t. $\sum_{i \in S} x_i \geq v(S) \forall S \neq \emptyset$ and $\sum_{i \in N} x_i = v(N)$.

- **Constant-sum:** $v(S) + v(N - S) = v(N) \forall S \subseteq N$
- **Essential:** $v(N) > \sum_{i \in N} v(\{i\})$
- Inessential $\implies v(S) = \sum_{i \in S} v(\{i\}) \implies$ const-sum
- Const-sum $\implies C(N, v) = \emptyset$ or $\{(v(\{i\}))_{i \in N}\}$
- Inessential $\implies C(N, v) = \{(v(\{i\}))_{i \in N}\}$
- Essential and const-sum $\implies$ empty core
- Games v and u are **strategically equivalent** if $\exists a > 0, c_1, \dots, c_n \in \mathbb{R}$ s.t. $u(S) = av(S) + \sum_{i \in S} c_i \forall S \subseteq N$
- if (N, u) and (N, v) are strategically equivalent:
 - u essential $\iff v$ essential
 - $x \in I(N, v) \iff ax + c \in I(N, u)$
 - $x \succ_S y$ in $v \iff ax + c \succ_S ay + c$ in u
 - $x \in C(N, v) \iff ax + c \in C(N, u)$
- char func $(0, 1)$ -reduced form: $v(\{i\}) = 0, v(N) = 1$
- $\forall$ essential game, $\exists$ reduced form game s-equiv to it. **Proof:** take $a = \frac{1}{v(N) - \sum_{i \in N} v(\{i\})}$, $c_i = -av(\{i\})$
- Classify small essential games in reduced form:
 1. 3-person const-sum: $v(\{i, j\}) = 1 \forall i \neq j$ and def
 2. 3-person: $\exists a, b, c \in [0, 1]$ s.t. $v(\{1, 2\}) = a$, $v(\{1, 3\}) = b$, $v(\{2, 3\}) = c$ and def
- **Shapley val:** $\phi(v)$ n -vector s.t.
 1. $\phi_i(v) \geq v(\{i\})$ (individual rational)
 2. $\sum_{i \in N} \phi_i(v) = v(N)$ (efficiency)
 3. $\phi_i(v) = \phi_j(v)$ if $v(S \cup \{i\}) = v(S \cup \{j\})$ for all S not containing i and j (symmetry)
 4. $\phi(v + w) = \phi(v) + \phi(w)$ (additive)
 5. $\phi_i(v) = 0$ if $v(S \cup \{i\}) = v(S)$ for all S (dummy)
- $\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{s!(n-s-1)!}{n!} (v(S \cup \{i\}) - v(S)) = \frac{1}{n} \sum_{s=0}^{n-1} \frac{1}{C_s^{n-1}} \sum_{S \subseteq N \setminus \{i\}, |S|=s} (v(S \cup \{i\}) - v(S))$
- If $v_A(S) = \alpha \mathbf{1}\{A \cup S \neq \emptyset\}$ for a fixed $A \subseteq N$, then $\phi_i(v_A) = \frac{\alpha}{|A|} \mathbf{1}\{i \in A\}$.